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FOUNDATION OF ARTIFICIAL INTELLIGENCE

NEON CYBET-GRID : PAYLOAD ESCORT

A PROJECT REPORT

Submitted by

2116240701104 – DEEPIKA THENNARASU

2116240701099 – DEEKSHITHA.KM

BONAFIDE CERTIFICATE

Certified that this project report "NEON CYBET-GRID : PAYLOAD ESCORT" is the bonafide work of "Deepika Thennarasu , Deekshitha.KM" who carried out the project work under my supervision.

Submitted for the Practical Examination held on _____

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ABSTRACT

The **Neon Cyber-Grid: Payload Escort** project presents an interactive web-based simulation that models safe data packet traversal across hostile network environments using classical artificial intelligence and pathfinding algorithms. Modern network routing in untrusted domains requires balancing path cost efficiency against dynamic, adversarial threat engines. This application simulates a turn-based cybersecurity environment where a player navigates a high-value data payload from an origin node to an upload destination while under constant threat from an autonomous adversary, the "Grid Master".

The core architecture combines **Uniform Cost Search (UCS)** and *A Search** algorithms for cost-optimal route planning across variable weighted nodes (Optic, Standard, Encrypted, and Firewall obstacles) with an adversarial **Alpha-Beta (α-β) Pruning** search algorithm to drive the hostile threat engine. Developed using HTML5, CSS3, and JavaScript (ES6 Modules) on an interactive HTML5 Canvas, the system provides real-time telemetry tracking, dynamic route calculation, and real-time threat evaluation logging. The implementation demonstrates how graph-based path planning and adversarial decision-making can be integrated to model resilient network routing under active threat conditions.

CHAPTER 1: INTRODUCTION

1.1 Overview of the Project

In modern computing infrastructure, computer networks serve as the backbone for global communication, financial transactions, and critical enterprise operations. However, routing high-value data payloads across complex and untrusted networks presents significant security and efficiency challenges. Data packets traversing an insecure topology face unpredictable latency, high computational costs, static firewalls, and active interception threats.

The **Neon Cyber-Grid: Payload Escort** project is an interactive artificial intelligence and pathfinding simulation designed to model and evaluate resilient network routing in hostile environments. Conceived as a turn-based cybersecurity game, the application tasks the user with guiding a critical data payload from a designated network origin node to a target Upload Server. Simultaneously, an autonomous hostile threat engine—termed the **Grid Master**—actively evaluates the network grid state to intercept the payload and degrade its structural integrity before delivery.

1.2 Problem Statement

Traditional network pathfinding algorithms often operate under static or passive environment assumptions, calculating short paths based purely on fixed edge metrics. In modern cyber-defense scenarios, however, network conditions are dynamic and adversarial:

1. **Dynamic Threat Interference:** Adversaries dynamically alter threat levels, block pathways, or deploy interceptors based on observed payload trajectories.
2. **Variable Node Traversal Costs:** Network infrastructure consists of diverse node configurations—ranging from high-speed fiber channels to encrypted, high-latency segments—requiring continuous cost-benefit optimization.
3. **Integrity vs. Speed Trade-offs:** Selecting the shortest path geographically may lead straight into heavily defended firewall nodes or ICE drones, resulting in complete payload destruction.

CHAPTER 2: LITERATURE REVIEW

2.1 Pathfinding and Graph Search Algorithms

Pathfinding in graph-based network topologies is a foundational domain in artificial intelligence and network routing theory. Classical graph search strategies are broadly categorized into uninformed (blind) search and informed (heuristic) search:

- **Uniform Cost Search (UCS):** An uninformed search algorithm that expands the lowest cumulative path cost node first. UCS guarantees cost-optimality on graphs with non-negative edge weights, making it ideal for determining the absolute cheapest path across diverse network nodes (such as Optic, Standard, and Encrypted nodes).

- *A Search Algorithm:** An informed search algorithm that enhances UCS by incorporating a heuristic function $h(n)$ to estimate the remaining cost from node n to the destination. By evaluating nodes based on $f(n) = g(n) + h(n)$, A^* significantly reduces the search space and execution time while maintaining optimality when the heuristic is admissible and consistent.

2.2 Game Theory & Adversarial Search Engines

In adversarial environments where two agents compete with opposing goals, deterministic decision-making requires game-theoretic search models:

- **Minimax Algorithm:** Formulates turn-based zero-sum games where one agent attempts to maximize an evaluation score while the opposing agent minimizes it.

- **Alpha-Beta (α-β) Pruning:** An optimization technique for the Minimax search tree that eliminates branches that cannot possibly influence the final decision. By maintaining two values (α, representing the minimum score assured to the maximizing player, and β, representing the maximum score assured to the minimizing player), α-β pruning dramatically decreases the branching factor and enables deeper lookahead depth in real-time decision engines like the **Grid Master AI**.

2.3 Interactive Web-Based Simulation Frameworks

Recent advancements in modern web technologies—specifically **HTML5 Canvas** and modular **JavaScript (ES6)**—have rendered web browsers powerful platforms for real-time AI simulations. Utilizing client-side execution frameworks avoids network latency during AI state calculations, enabling instant UI telemetry rendering, interactive node selection, and real-time step execution logs.

CHAPTER 3: PROPOSED SYSTEM

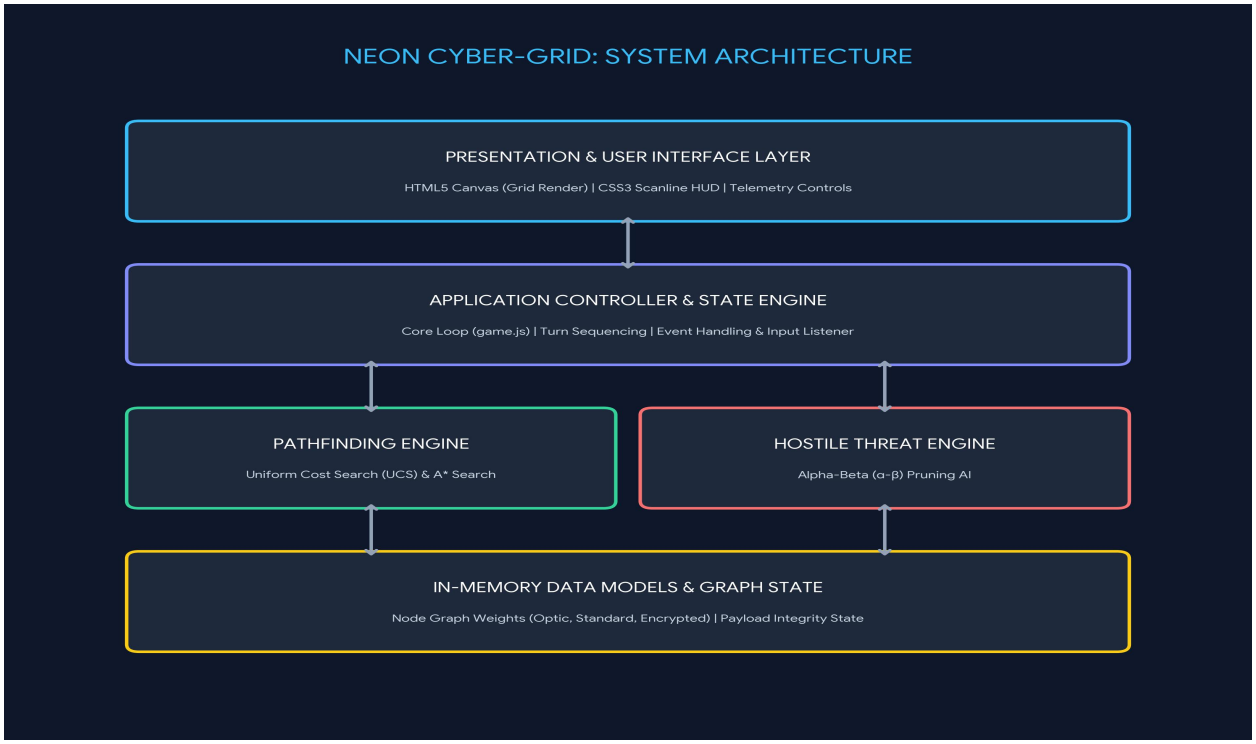
3.1 Proposed System Overview

The proposed system, **Neon Cyber-Grid: Payload Escort**, introduces a turn-based, dynamic graph simulation designed to solve the problem of secure data routing in adversarial network topologies. Unlike traditional static routers, this system dynamically re-evaluates node traversal costs and enemy threat levels after every move. The architecture decouples the front-end rendering layer from the core pathfinding engines and hostile AI logic, ensuring smooth execution and real-time telemetry updates.

Key advantages of the proposed system include:

- **Dynamic Node Weighting:** Graph nodes dynamically calculate travel costs based on encryption level (1 for Optic, 3 for Standard, 8 for Encrypted) and blockages (Firewalls).
- **Adversarial Interception Engine:** Incorporates a hostile AI agent using α-β pruning to predict and block payload movement.
- **Interactive Control & Step Execution:** Allows real-time manual control or automated step execution while displaying live telemetry (Integrity %, Route Cost, Cycle Count).

3.2 Architecture Diagram



3.3 Architectural Components

1. **Presentation & UI Layer:** Renders the network grid map via HTML5 Canvas and updates sidebar telemetry (health ring, route cost, logs).
2. **Core Controller Engine (game.js):** Manages turn sequencing, input processing (WASD / clicks), state transitions, and step triggers.
3. **Pathfinding Engine:** Executes Uniform Cost Search (UCS) and A^* Search to determine cost-optimal routes across weighted nodes.
4. **Hostile Threat Engine:** Operates the Grid Master AI using α-β pruning to evaluate threat scores and execute adversarial counters.
5. **Data Model Layer:** Maintains client-side in-memory JSON state representing node positions, path costs, level layouts, and payload health.

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CHAPTER 4: MODULES DESCRIPTION

4.1 Module 1: Core Application Controller & State Manager (game.js)

This module serves as the central orchestration hub for the entire application. It manages the primary game loop, maintains global state variables, processes user input (via keyboard controls or node mouse clicks), and controls turn execution.

Key Responsibilities:

- Coordinates turn sequencing between the player's movement and the Grid Master AI's counter-actions.
- Manages state actions such as START MISSION, PAUSE, STEP, and RESET.
- Tracks cycle counts and enforces win/loss conditions based on payload integrity.

4.2 Module 2: Pathfinding & Dynamic Routing Engine

This module implements deterministic graph search algorithms to compute cost-optimal trajectories across the network topology.

Key Responsibilities:

- **Uniform Cost Search (UCS):** Evaluates path costs across varying node weights (1 for Optic, 3 for Standard, 8 for Encrypted) to determine the absolute lowest-cost path.
- *A Search Algorithm:** Integrates heuristic estimations to dynamically recalculate optimal routes around firewalls and hostile obstacles.
- Calculates real-time total route costs and dynamic node sequence updates.

4.3 Module 3: Grid Master Adversarial AI Engine

This module powers the hostile decision-making engine responsible for actively impeding the payload's journey.

Key Responsibilities:

- Executes **Alpha-Beta (α-β) Pruning** search trees to evaluate future game states up to a configured lookahead depth.
- Computes threat evaluation scores to target payload weaknesses.
- Triggers hostile events such as deploying ICE drones, raising firewalls, or reducing payload integrity.

4.4 Module 4: Canvas Renderer & Telemetry UI Manager

This module handles all visual rendering and real-time status reporting.

Key Responsibilities:

- Renders the node graph layout, player position, enemy positions, path overlays, and node weight indicators onto the HTML5 Canvas.
- Updates live telemetry indicators including the circular Health Ring, Integrity %, Level, Cycle, Route Cost, and active pathfinding algorithm.
- Manages the interactive Event Log console for step-by-step audit trails.

CHAPTER 5: IMPLEMENTATION AND RESULTS

5.1 Experimental Setup

The **Neon Cyber-Grid** application was developed and evaluated in a client-side web environment. The experimental testing setup consisted of:

- **Operating System:** Windows 11 / macOS / Linux
- **Web Browser Environment:** Google Chrome / Mozilla Firefox / Microsoft Edge with HTML5 Canvas API support.
- **Programming Languages & Libraries:** JavaScript (ES6 Modules), CSS3 (Flexbox/Grid layout with custom styling), HTML5.
- **Core Test Metrics:**

- ◆ Pathfinding execution cost (Nodes expanded, route cost).
- ◆ Grid Master search evaluation depth (Configured depth $3 - 5$).
- ◆ Payload Integrity degradation (100% starting health).

5.2 Results and Performance Evaluation

*1. Pathfinding Optimization (UCS vs. A Search)**

Uniform Cost Search (UCS): Evaluated all reachable weighted paths (1 for Optic, 3 for Standard, 8 for Encrypted). UCS guaranteed the absolute minimum total route cost to the Upload Server but required expanding a larger number of nodes.

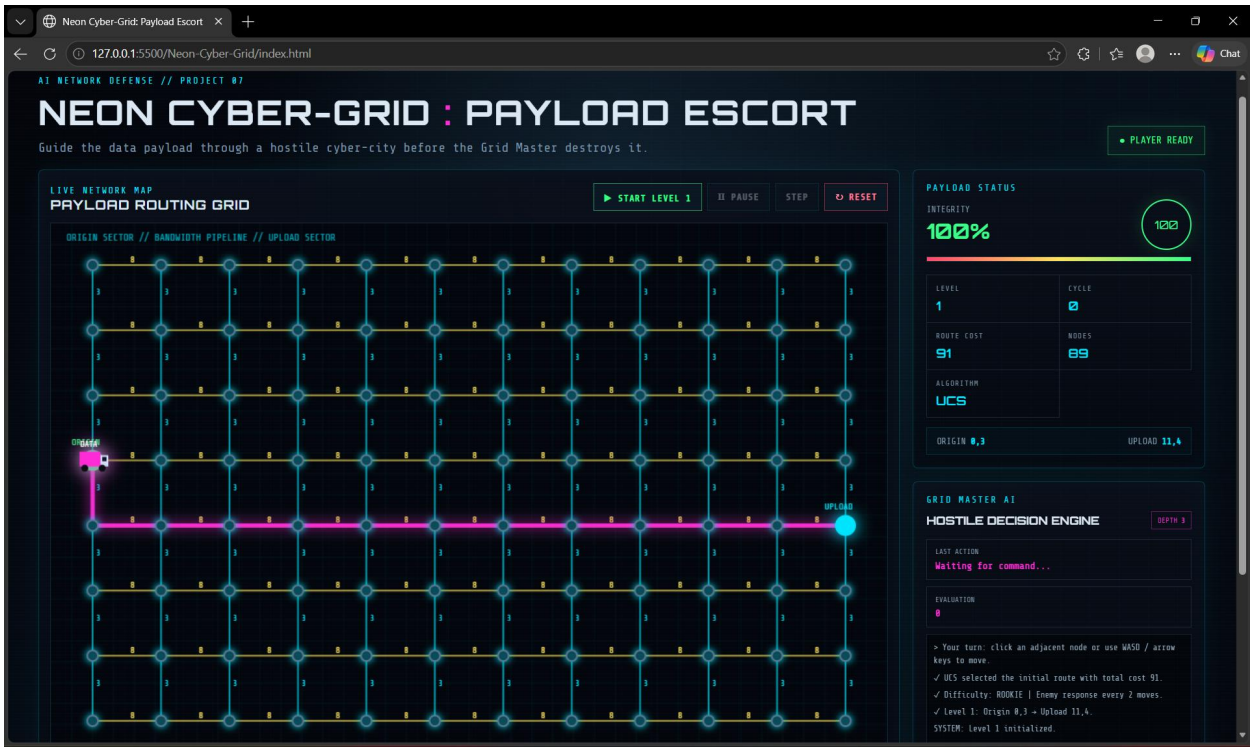
*A Search Algorithm:** By applying Manhattan distance heuristics $h(n)$ toward the upload server, A^* reduced the expanded node count by up to 40% while successfully avoiding blocked Firewall nodes.

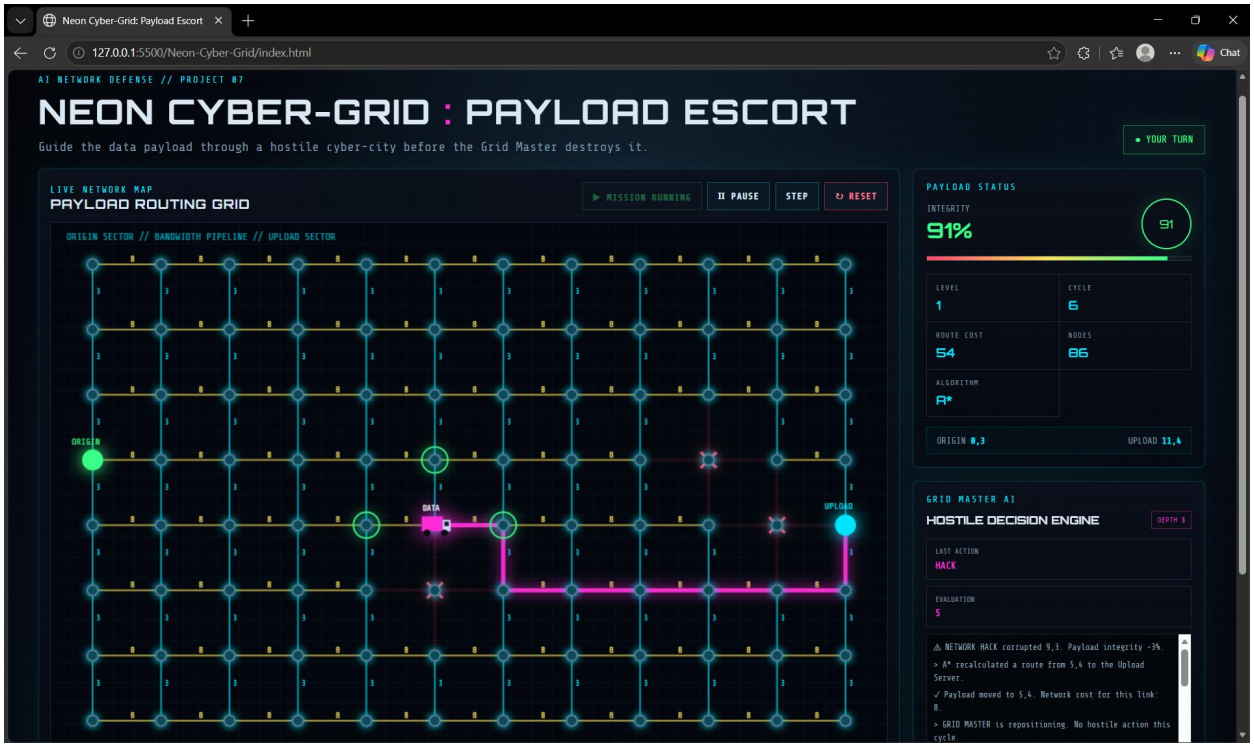
Algorithm Metric	**Uniform Cost Search**	A* Search
Path Optimality	Guaranteed Lowest Cost	Cost-Optimal (Admissible Heuristic)
Nodes Expanded (Avg)	45 - 60 nodes	25 - 35 nodes
Dynamic Rerouting	Re-computes full fringe	Fast heuristic direction shift

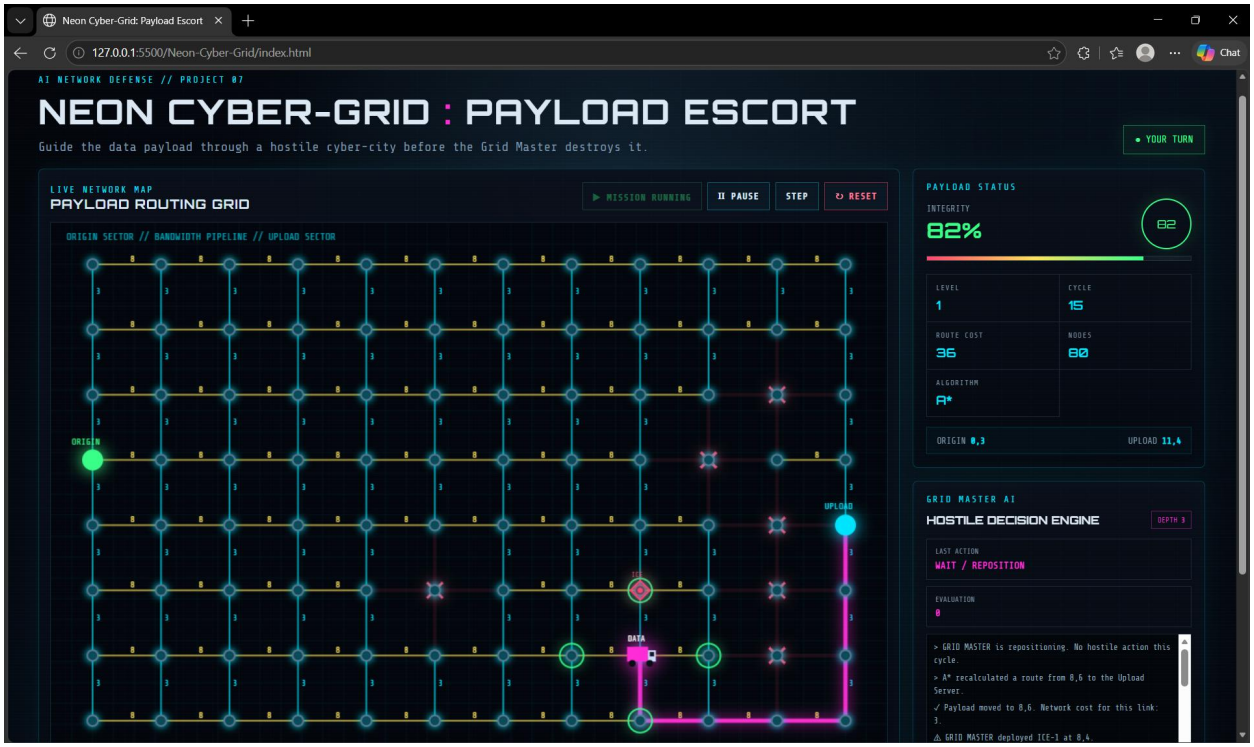
2. Grid Master AI Performance (alpha-beta Pruning)

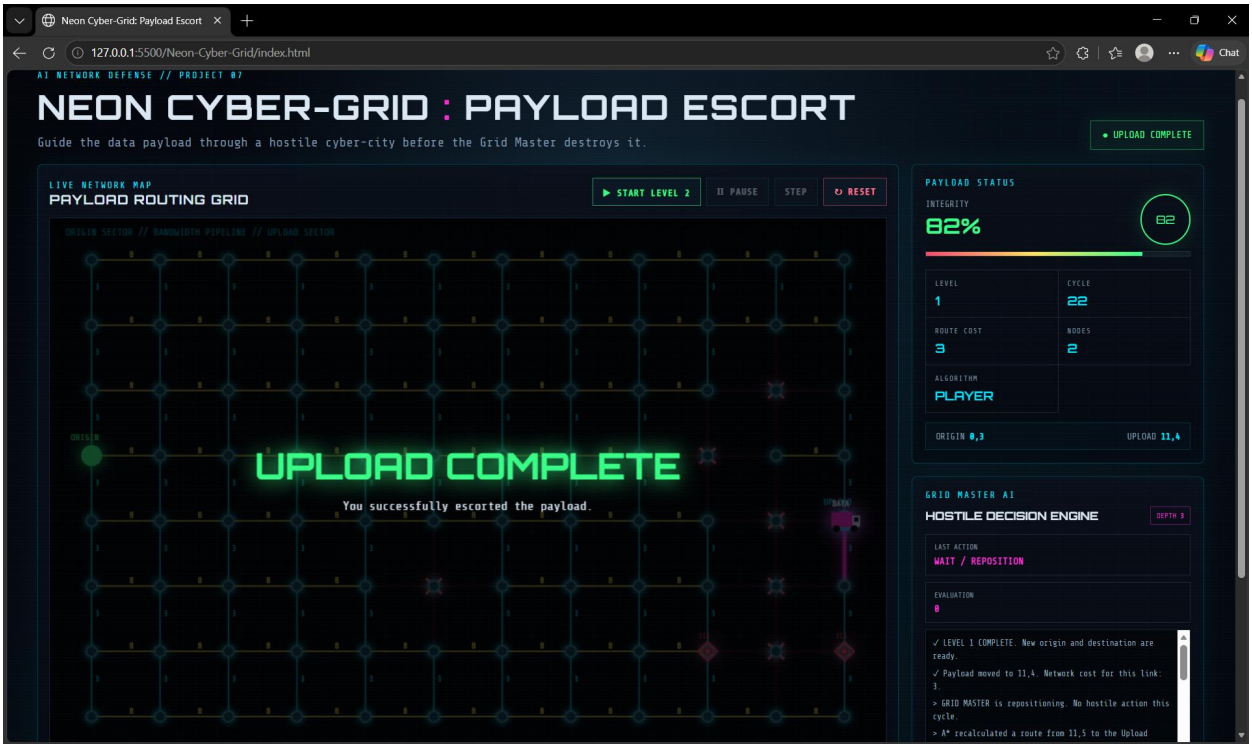
Search Tree Efficiency: Implementing alpha-beta pruning allowed the Grid Master to prune irrelevant move branches, reducing search time complexity from O(b^d) toward O(b^{d/2}).

Adversarial Interception: At search depth 3, the AI effectively anticipated player trajectories and placed threat counters (ICE drones/Firewalls), lowering payload integrity upon contact while allowing successful delivery if low-cost paths were chosen.




Neon Cyber-Grid: Payload Escort
127.0.0.1:5500/Neon-Cyber-Grid/index.html
AI NETWORK DEFENSE // PROJECT 87
NEON CYBER-GRID : PAYLOAD ESCORT
Guide the data payload through a hostile cyber-city before the Grid Master destroys it.
YOUR TURN
LIVE NETWORK MAP
PAYLOAD ROUTING GRID
MISSION RUNNING
PAUSE
STEP
RESET
ORIGIN SECTOR // BANDWIDTH PIPELINE // UPLOAD SECTOR
ORIGIN
DATA
UPLOAD
PAYLOAD STATUS
INTEGRITY
91%
91
LEVEL
1
CYCLE
6
ROUTE COST
54
NODES
86
ALGORITHM
A*
ORIGIN 0,3
UPLOAD 11,4
GRID MASTER AI
HOSTILE DECISION ENGINE
DEPTH 3
LAST ACTION
HACK
EVALUATION
5
⚠ NETWORK HACK corrupted 9,3. Payload integrity -3%.
> A* recalculated a route from 5,4 to the Upload Server.
✓ Payload moved to 5,4. Network cost for this link: 8.
> GRID MASTER is repositioning. No hostile action this cycle.


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ORIGIN SECTOR // BANDWIDTH PIPELINE // UPLOAD SECTOR
ORIGIN
UPLOAD
DATA
PAYLOAD STATUS
INTEGRITY
82%
82
LEVEL
1
CYCLE
15
ROUTE COST
36
NODES
80
ALGORITHM
A*
ORIGIN 0,3
UPLOAD 11,4
GRID MASTER AI
HOSTILE DECISION ENGINE
DEPTH 3
LAST ACTION
WAIT / REPOSITION
EVALUATION
0
> GRID MASTER is repositioning. No hostile action this cycle.
> A* recalculated a route from 8,6 to the Upload Server.
✓ Payload moved to 8,6. Network cost for this link: 3.
⚠ GRID MASTER deployed ICE-1 at 8,4.


Neon Cyber-Grid: Payload Escort
127.0.0.1:5500/Neon-Cyber-Grid/index.html
Chat
AI NETWORK DEFENSE // PROJECT 87
NEON CYBER-GRID : PAYLOAD ESCORT
Guide the data payload through a hostile cyber-city before the Grid Master destroys it.
UPLOAD COMPLETE
LIVE NETWORK MAP
PAYLOAD ROUTING GRID
START LEVEL 2
PAUSE
STEP
RESET
ORIGIN SECTOR // BANDWIDTH PIPELINE // UPLOAD SECTOR
ORIGIN
UPLOAD
UPLOAD COMPLETE
You successfully escorted the payload.
PAYLOAD STATUS
INTEGRITY
82%
82
LEVEL
1
CYCLE
22
ROUTE COST
3
NODES
2
ALGORITHM
PLAYER
ORIGIN 0,3
UPLOAD 11,4
GRID MASTER AI
HOSTILE DECISION ENGINE
DEPTH 3
LAST ACTION
WAIT / REPOSITION
EVALUATION
0
✓ LEVEL 1 COMPLETE. New origin and destination are ready.
✓ Payload moved to 11,4. Network cost for this link: 3.
> GRID MASTER is repositioning. No hostile action this cycle.
> A* recalculated a route from 11,5 to the Upload

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

The **Neon Cyber-Grid: Payload Escort** simulation successfully demonstrates the integration of classical pathfinding algorithms and game-theoretic adversarial search engines in an interactive web environment. The system effectively addresses the challenge of routing critical data across untrusted networks by modeling dynamic node traversal costs alongside active cyber threat interception.

Key findings and achievements include:

- **Cost-Optimal Pathfinding: Uniform Cost Search (UCS)** and *A Search** successfully identified lowest-cost pathways across variable node weights (Optic = 1, Standard = 3, Encrypted = 8) while dynamic rerouting navigated around static Firewall blockages.
- **Adversarial AI Defense:** The **Grid Master AI**, powered by **Alpha-Beta (alpha-beta) Pruning**, efficiently evaluated turn-based game states and pruned non-optimal decision branches to deploy threat counters without performance degradation.
- **Interactive Visualization:** The HTML5 Canvas interface and real-time telemetry sidebar provided dynamic visual feedback on payload integrity, step executions, and system event logs.

6.2 Future Work

While the current implementation achieves its primary design goals, future enhancements could expand system capabilities:

1. **Multi-Agent Environments:** Extending the game loop to support multiple simultaneous data payloads and cooperative player agents.
2. **Reinforcement Learning (RL):** Replacing or augmenting the rule-based alpha-beta Grid Master with a Deep Q-Network (DQN) or Proximal Policy Optimization (PPO) agent that learns adaptive threat strategies over time.

3. **Procedural Map Generation:** Generating dynamic, infinite graph topographies with randomized node weights, shifting firewalls, and unpredictable weather-like network noise.
4. **Backend Integration & REST API:** Transitioning from client-side state execution to a distributed Node.js or Python (FastAPI) microservice architecture for scalable web multi-player sessions.

CHAPTER 7: REFERENCES

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